TRIAGE

Diagnose before you retry

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LLM

Framework

Four levels, cheap to expensive

LEVEL 0

Hybrid retrieval

Sparse and dense search, one answer from the target model

LEVEL 1

Answer sampling

Draw a few answers and see whether they agree

LEVEL 2

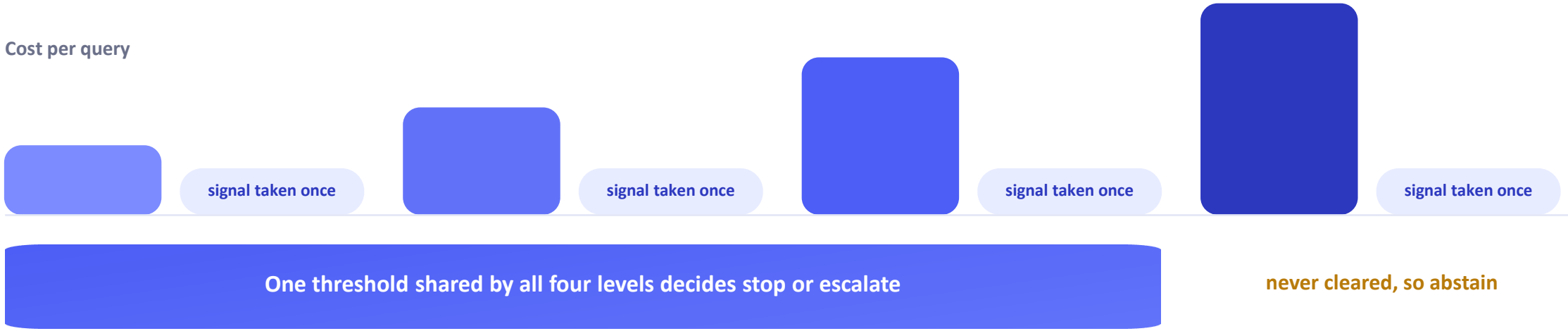
Recovery operator

Run the single repair that matches the predicted cause

LEVEL 3

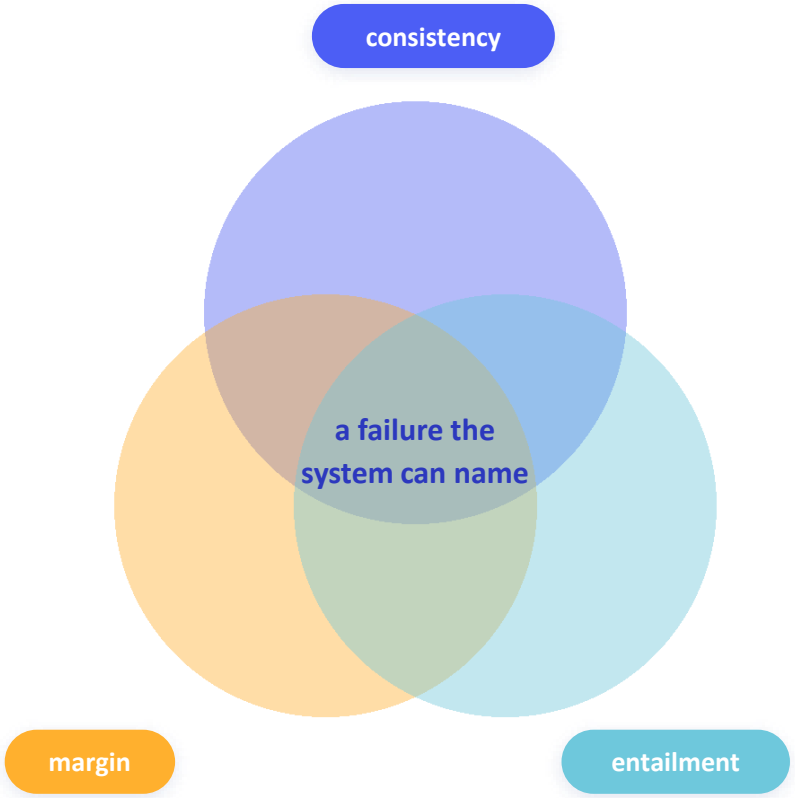
Reasoning model

A large model runs multi round retrieval as a last attempt



Signals

Three cheap measurements, three blind spots



Answer consistency

generation side

Sample several answers and group the ones that mean the same thing. A large dominant group means the model keeps landing on one answer.

Retrieval margin

retrieval side

The score gap between the best and the weakest retrieved document. A flat gap means the search never concentrated on anything.

Evidence entailment

grounding side

Whether a small set of documents actually implies the answer, checked by a light inference model rather than by the generator itself.

Diagnosis

Five failure types, five responses

A type is not a description of a cause. It is the name of the repair that was shown to work on that query.

TYPE 1



Over constrained

split by constraint

The question bundles several independent conditions. Split it into atomic sub questions, search for each in parallel, then keep only the candidates that satisfy every condition.

TYPE 2



Compositional

retrieve in order

A later step depends on the answer of an earlier one. Answer the first sub question, place that answer into the next one, and search again.

TYPE 3



Vocabulary mismatch

paraphrase and search

The wording of the question does not match the wording of the corpus. Rewrite it in encyclopedia style, expand abbreviations, and lean on the sparse retriever.

TYPE 4



Entity ambiguity

resolve the entity

The mention could point to more than one real entity. Choose the consistent one from the candidate titles, then search again with the resolved name.

TYPE 5



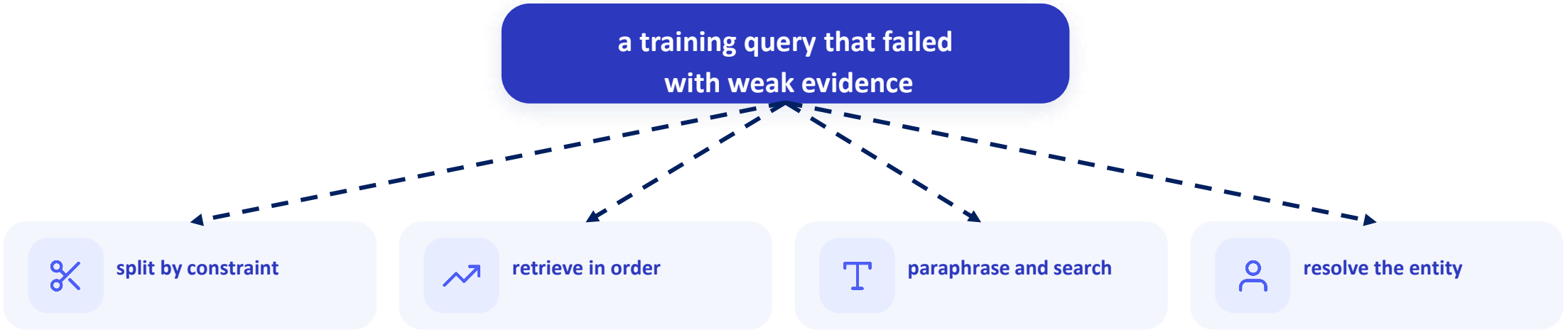
No recovery

skip to level 3

No operator was ever confirmed to fix this query. It goes straight to the large reasoning model instead of being abandoned on the spot.

Labeling

The label comes from execution, not from annotation



All four run from the same starting point, so no operator inherits the retrieval another one changed.

recovered means entailment rises above the threshold and the gold answer is matched

Exactly one operator recovers it
that operator names the failure type of this query

No operator recovers it
the query is labelled no recovery and later skips level 2

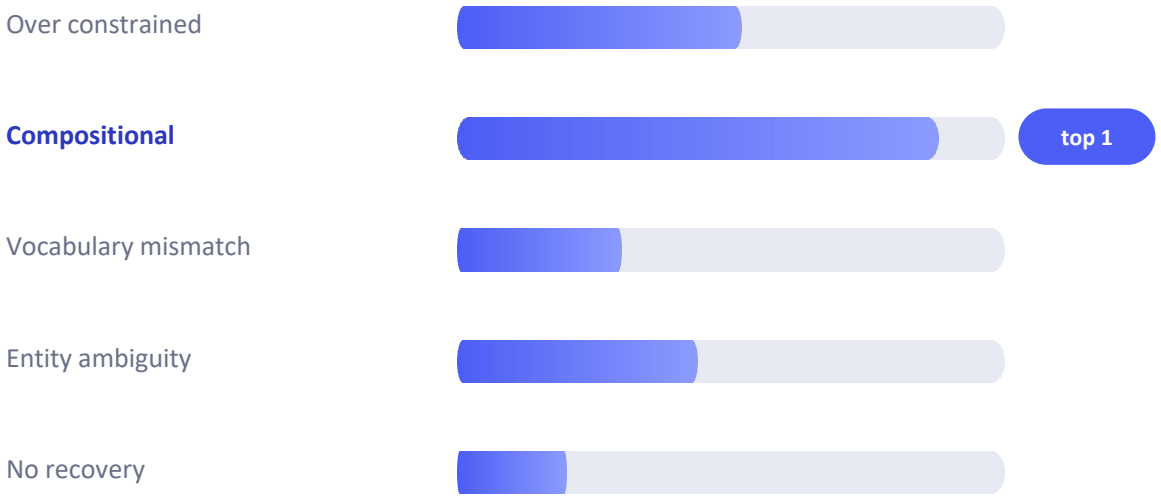
Routing

A tiny model picks the repair

The diagnoser

A multinomial logistic regression reads the signal vector and returns one probability per failure type. Nothing heavier is used.

PROBABILITY OVER THE FIVE TYPES



Why a linear model

Labelled failures are few, so a larger model would memorise them. The coefficients can also be read to see which signal separates which type.

Only the top prediction runs

One operator is executed rather than four. If the prediction is wrong, the upper level is still there to absorb the query.

Signals are measured again

After the operator runs, the signals are recomputed, so the stopping decision always looks at the current answer.

The cost of deciding

At inference the diagnoser only predicts, so beyond the signal extraction already paid for it costs effectively nothing.

Stopping

Turning a threshold into a promise

01

Per level estimate

At each level a small model turns the signal vector into a probability that the current answer is correct. The link between signals and correctness differs by level, so each level keeps its own estimator.

02

One shared threshold

Because every level now speaks in probability, a single number can be compared everywhere. The pipeline stops at the first level whose score clears it.

03

Calibrate that number

Conformal risk control picks the threshold on a held out calibration set, with a correction for how many samples that set holds.

What the chosen level buys you



the operator sets this level, the system honours it

If nothing clears the threshold

the query is not answered, and abstaining is treated as no loss, which is safer than a confident wrong answer

So abstention is capped

abstaining on everything would satisfy the target trivially, so the controlled quantity is the risk over answered queries

Cost

Paying for a decision only when it saves more

How cost is counted

Every level a query passed through is added up, not just the level where it stopped, so escalation never looks cheaper than it is. Search calls and inference calls are converted into one common unit.

The adaptive probe

Level 1 draws two answers first. If they entail each other the query stops there, and only diverging queries draw more samples up to a cap, so easy queries finish early.

When the probe is worth running

what the probe costs



the extra samples and entailment checks paid at level 1 for a query that did not stop

less than

what the probe avoids



the expensive upper levels that never run because the query was allowed to finish early

Experiment Setup

How the comparison is kept fair

Datasets

- Natural Questions
- TriviaQA
- PopQA
- AmbigQA
- 2WikiMultiHopQA
- MuSiQue

Single hop, long tail, ambiguous and multi hop queries, plus unanswerable items for the abstention check.

Unified harness

- one corpus
- BM25 and bge-m3
- Qwen2.5-7B-Instruct
- DeBERTa NLI
- QwQ-32B
- one answer format

Corpus, retriever, generator and answer format are held constant so only the control logic varies.

Baselines

- fixed pipelines
- CRAG
- Adaptive-RAG
- FLARE
- DRAGIN
- SeaKR
- SKR
- Probing-RAG

Reported numbers are not cited, every method is reproduced inside the same harness.

Metrics

- exact match
- token level F1
- cost per query
- realized error rate
- response rate

Accuracy, cost and reliability are read together, with the calibration split repeated many times.

Summary

What the framework adds



Failure types defined by execution

A type is assigned by running every operator on a training query and keeping the one that both restores the evidence and reaches the gold answer, so the label can be verified rather than argued about.



The repair follows the diagnosis

The pipeline switches the operator according to the predicted cause, which beats fixing one operator or picking at random, and costs far less than running all of them.



A stopping rule with a target

Per level probability estimates plus one shared threshold hold the error rate of returned answers under a level the operator chooses, even though escalation changes the answer itself.